

Lecture 3

Kinematics of particles II : Curvilinear Motion

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MC2103
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H

<https://youtu.be/3fFWPCQpggQ>

3 Kinematics of particles II: Curvilinear Motion

Curvilinear Motion: Position, Velocity & Acceleration

The snowboarder and the train both undergo curvilinear motion.

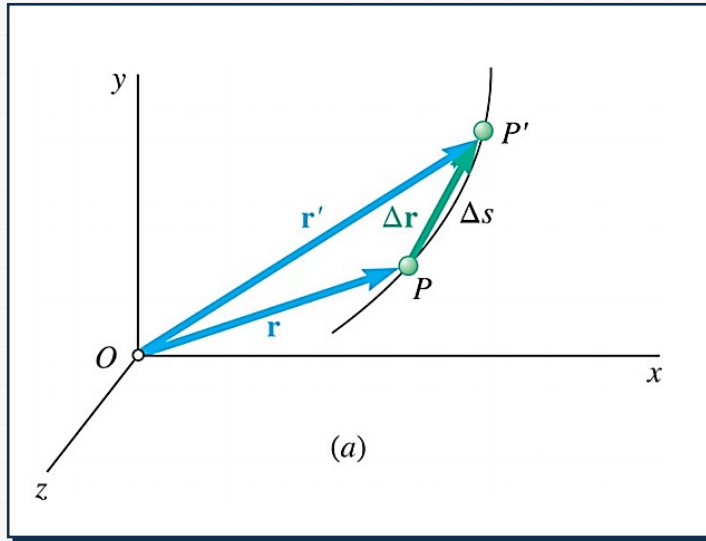
➡ A particle moving along a curve other than a straight line is in **curvilinear motion**.



3 Kinematics of particles II: Curvilinear Motion

Curvilinear Motion: Position, Velocity & Acceleration

⚙️ Position vector

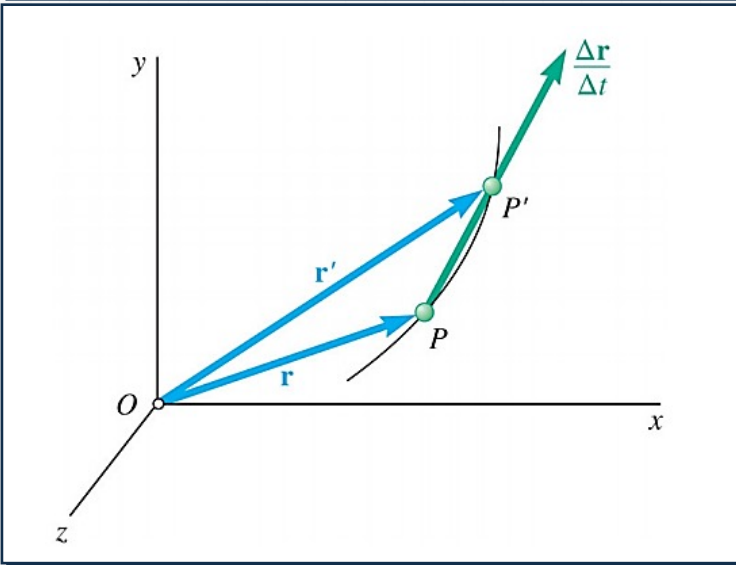


- The **position vector** of a particle at time t is defined by a vector between origin O of a fixed reference frame and the position occupied by particle.
- Consider a particle which occupies position P defined by $\vec{r}$ at time t and P' defined by $\vec{r}'$ at $t + \Delta t$

3 Kinematics of particles II: Curvilinear Motion

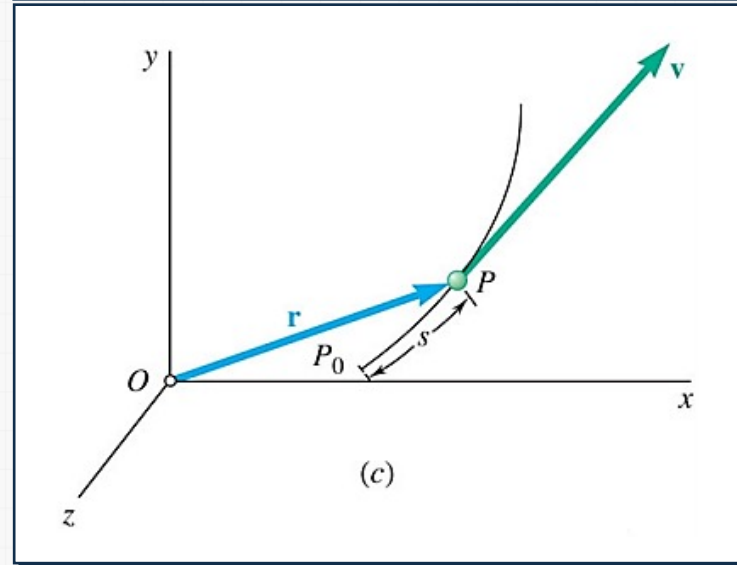
Curvilinear Motion: Position, Velocity & Acceleration

Instantaneous velocity
(vector)



$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

Instantaneous speed
(scalar)



$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

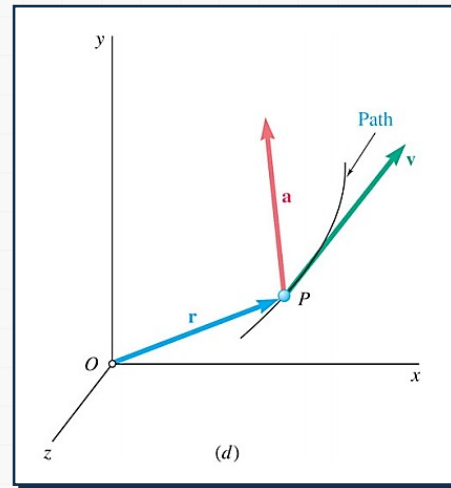
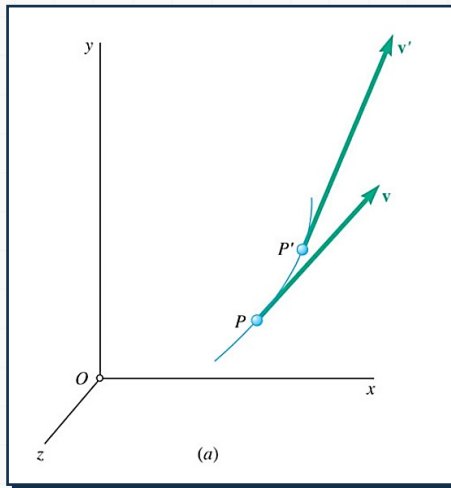
3 Kinematics of particles II: Curvilinear Motion

Curvilinear Motion: Position, Velocity & Acceleration

⚙️ Velocity

- Consider velocity $\vec{v}$ of a particle at time t and velocity $\vec{v}'$ at, $t + \Delta t$

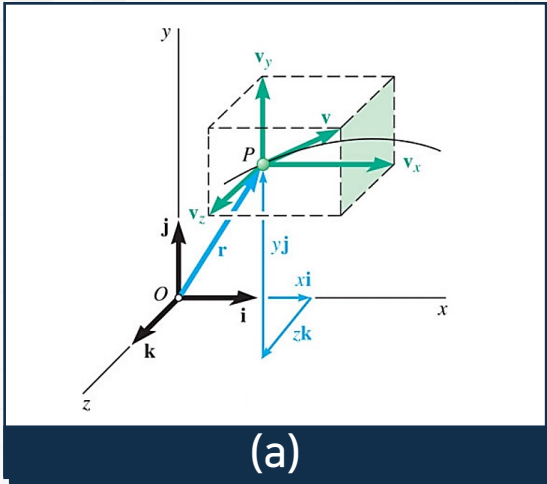
$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \text{instantaneous acceleration (vector)}$$



- In general, the acceleration vector is not tangent to the particle path and velocity vector.

3 Kinematics of particles II: Curvilinear Motion

Rectangular Components of Velocity & Acceleration



- When position vector of particle P is given by its rectangular components,

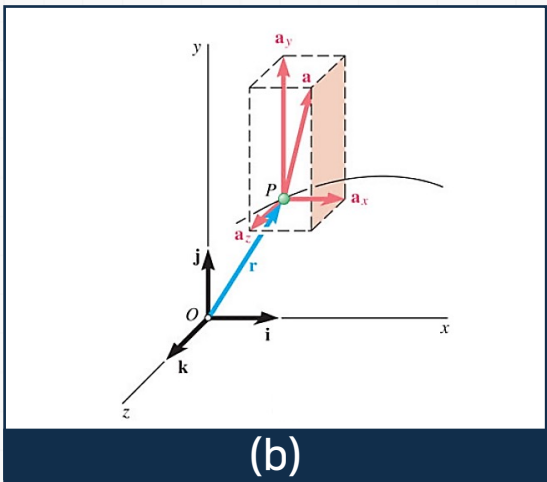
$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

- Velocity vector,

$$\begin{aligned}\vec{v} &= \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k} = \dot{x}\vec{i} + \dot{y}\vec{j} + \dot{z}\vec{k} \\ &= v_x\vec{i} + v_y\vec{j} + v_z\vec{k}\end{aligned}$$

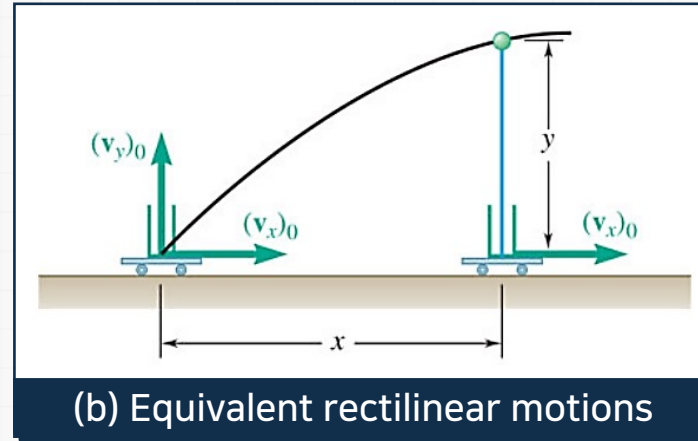
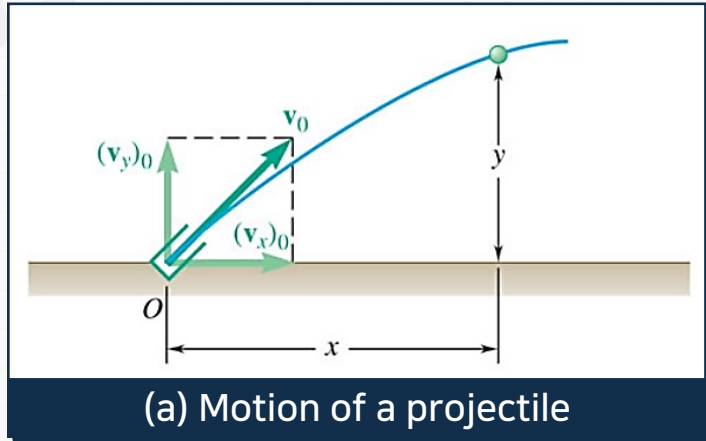
- Acceleration vector,

$$\begin{aligned}\vec{a} &= \frac{d^2x}{dt^2}\vec{i} + \frac{d^2y}{dt^2}\vec{j} + \frac{d^2z}{dt^2}\vec{k} = \ddot{x}\vec{i} + \ddot{y}\vec{j} + \ddot{z}\vec{k} \\ &= a_x\vec{i} + a_y\vec{j} + a_z\vec{k}\end{aligned}$$



3 Kinematics of particles II: Curvilinear Motion

Rectangular Components of Velocity & Acceleration



- Rectangular components are particularly effective when component accelerations can be integrated independently, example: motion of a projectile,

$$a_x = \ddot{x} = 0$$

$$a_y = \ddot{y} = -g$$

$$a_z = \ddot{z} = 0$$

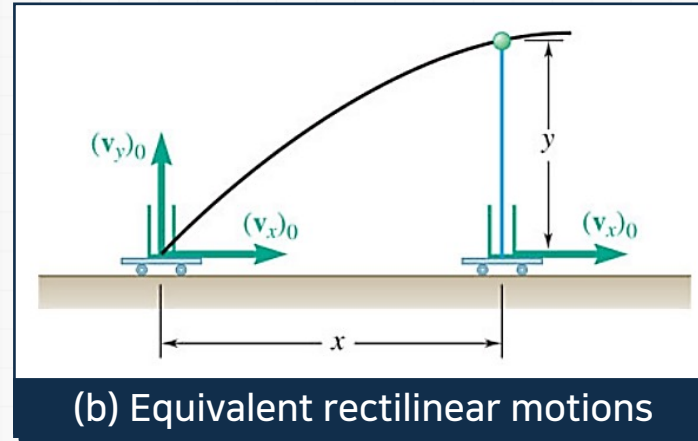
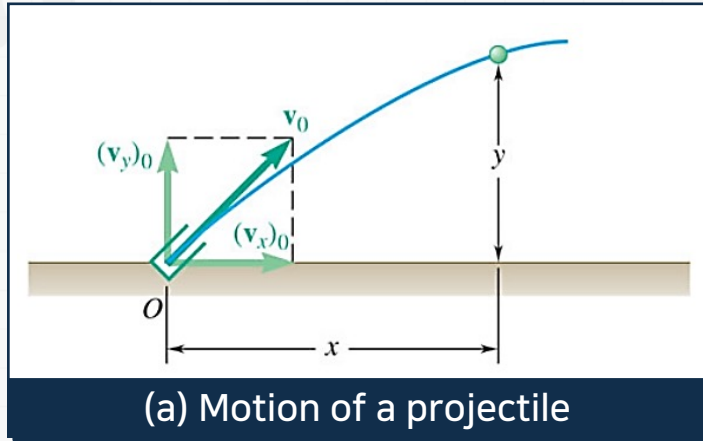
with initial conditions,

$$x_0 = y_0 = z_0 = 0$$

$$(v_x)_0, (v_y)_0, (v_z)_0 = 0$$

3 Kinematics of particles II: Curvilinear Motion

Rectangular Components of Velocity & Acceleration

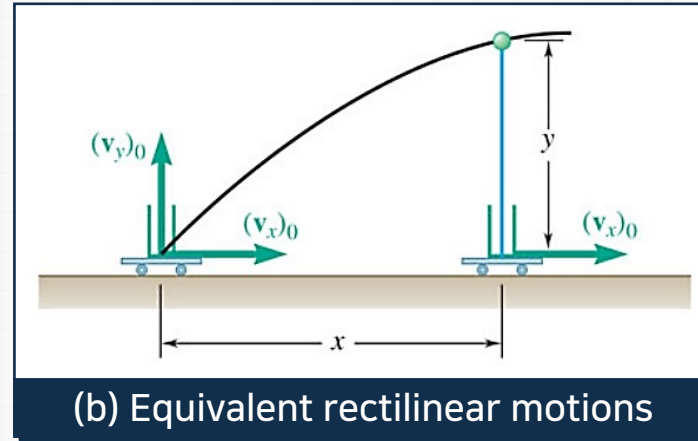
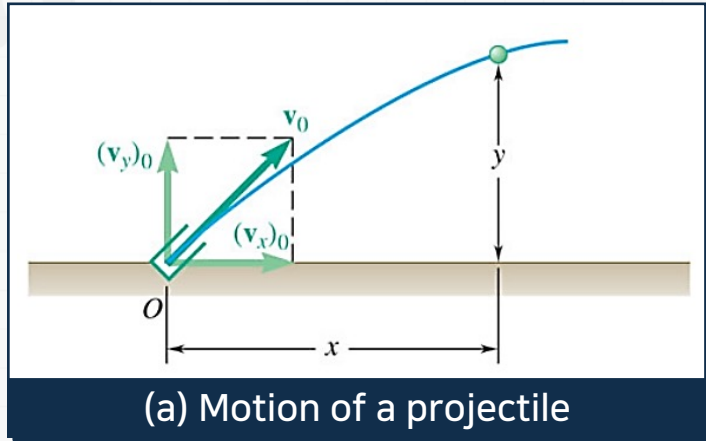


Integrating twice yields

$$\begin{array}{lll} v_x = (v_x)_0 & v_y = (v_y)_0 - gt & v_z = 0 \\ x = (v_x)_0 t & y = (v_y)_0 t - \frac{1}{2}gt^2 & z = 0 \end{array}$$

3 Kinematics of particles II: Curvilinear Motion

Rectangular Components of Velocity & Acceleration



- Motion in horizontal direction is uniform.
- Motion in vertical direction is uniformly accelerated.
- Motion of projectile could be replaced by two independent rectilinear motions.

3 Kinematics of particles II: Curvilinear Motion

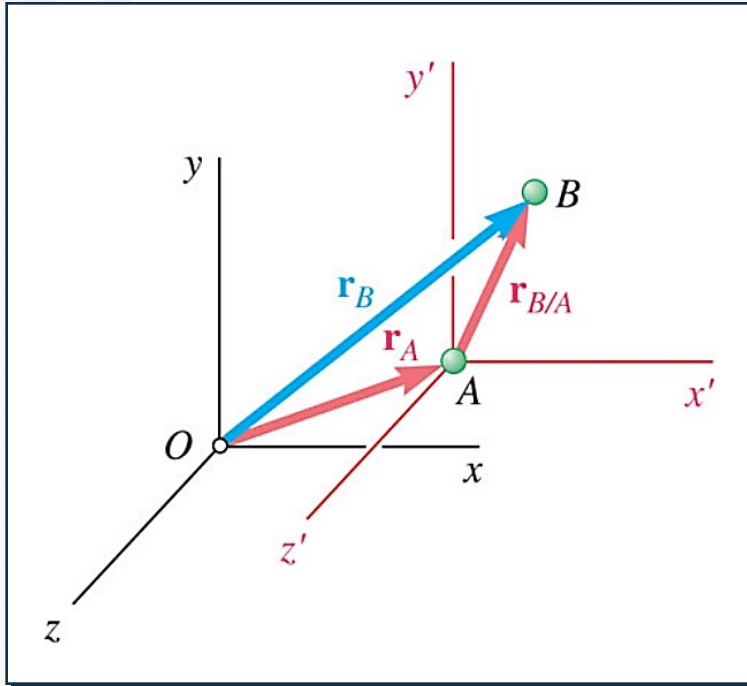
Motion Relative to a Frame in Translation

It is critical for a pilot to know the relative motion of his helicopter with respect to the aircraft carrier to make a safe landing.



3 Kinematics of particles II: Curvilinear Motion

Motion Relative to a Frame in Translation

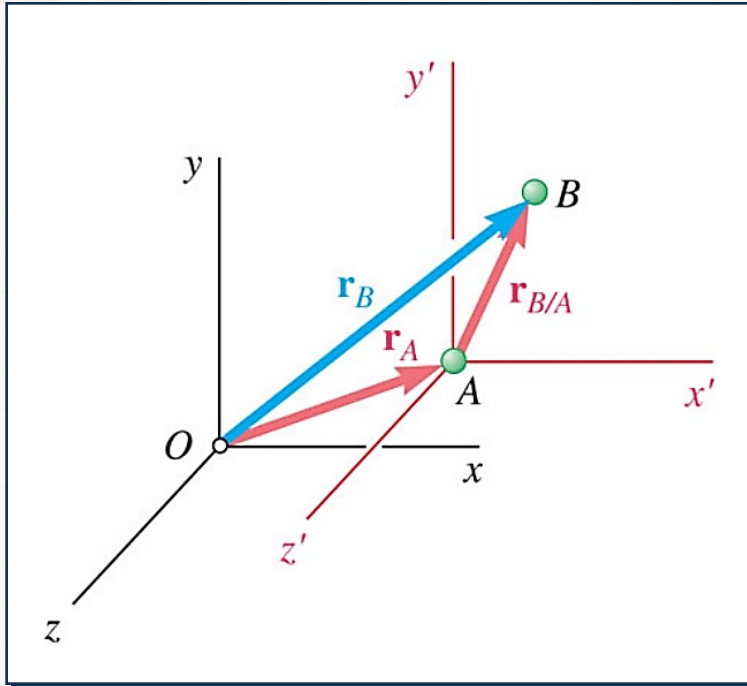


- Designate one frame as the *fixed frame of reference*. All other frames not rigidly attached to the fixed reference frame are *moving frames of reference*.

- Position vectors for A and B with respect to the fixed frame of reference $Oxyz$ are $\vec{r}_A$ and $\vec{r}_B$.

3 Kinematics of particles II: Curvilinear Motion

Motion Relative to a Frame in Translation



- Vector $\vec{r}_{B/A}$ joining A and B defines the position of B with respect to the moving frame $Ax'y'z'$ and
$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A} .$$

- Differentiating twice,

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A}$$

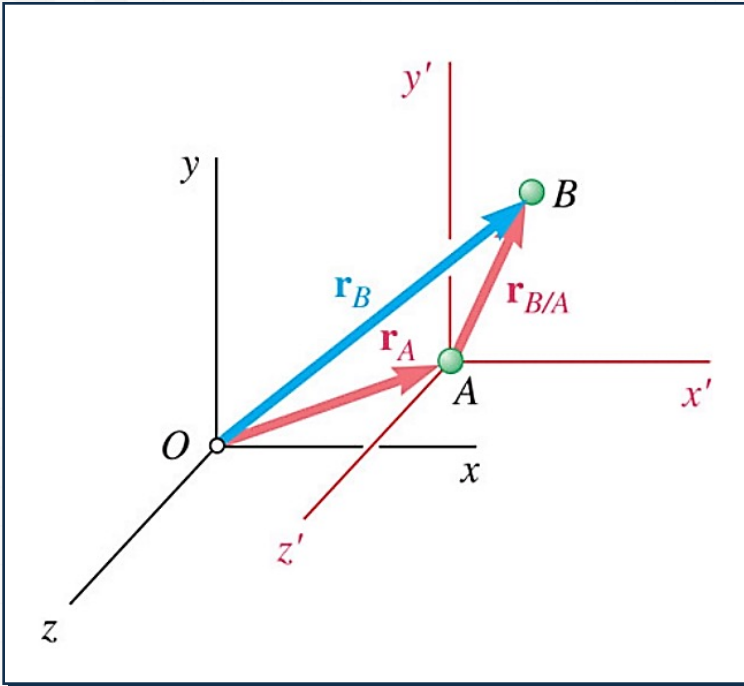
$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

$\vec{v}_{B/A}$ = velocity of B relative to A .

$\vec{a}_{B/A}$ = acceleration of B relative to A .

3 Kinematics of particles II: Curvilinear Motion

Motion Relative to a Frame in Translation



- Absolute motion of B can be obtained by combining motion of A with relative motion of B with respect to moving reference frame attached to A .

3 Kinematics of particles II: Curvilinear Motion

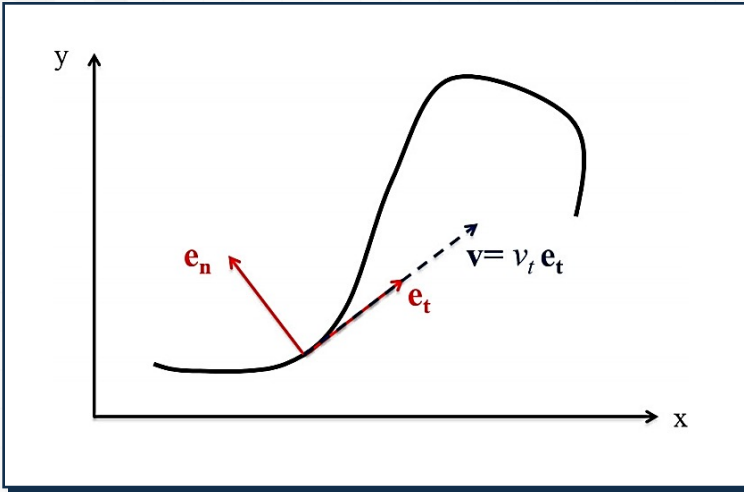
Tangential and Normal Components

If we have an idea of the path of a vehicle or object, it is often convenient to analyze the motion using tangential and normal components (sometimes called *path* coordinates).



3 Kinematics of particles II: Curvilinear Motion

Tangential and Normal Components



ρ = the instantaneous radius of curvature

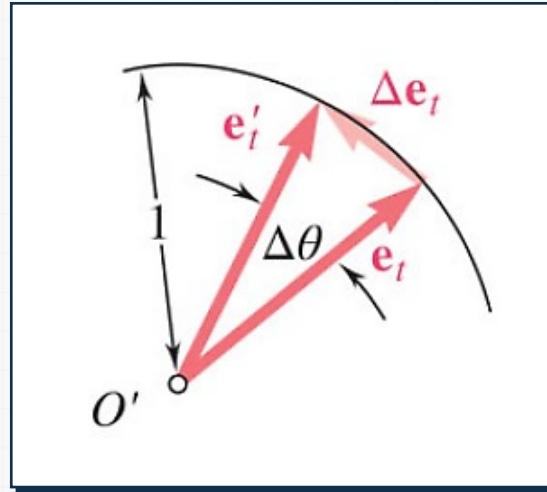
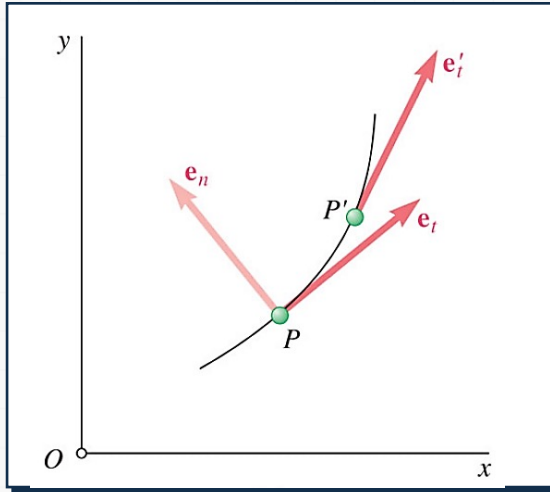
$$\vec{v} = v \hat{e}_t$$

$$\vec{a} = \frac{dv}{dt} \hat{e}_t + \frac{v^2}{\rho} \hat{e}_n$$

- The tangential direction ($\hat{e}_t$) is tangent to the path of the particle. The velocity vector of a particle is in this direction.
- The normal direction ($\hat{e}_n$) is perpendicular to $\hat{e}_t$ and points towards the inside of the curve.
- The acceleration can have components in both the $\hat{e}_n$ and $\hat{e}_t$ directions.

3 Kinematics of particles II: Curvilinear Motion

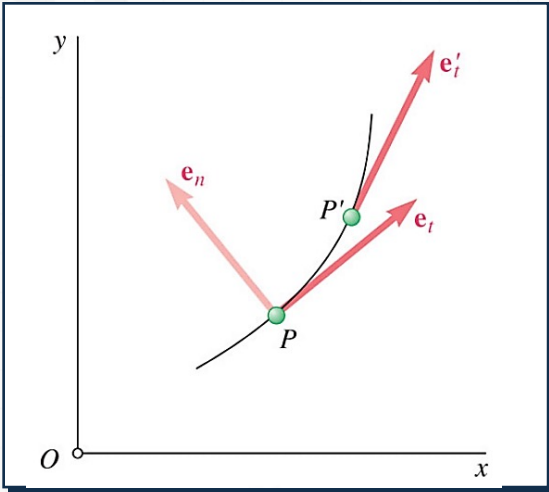
Tangential and Normal Components



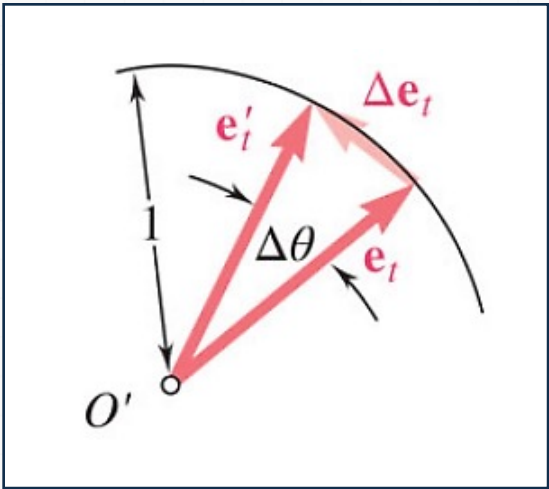
- To derive the acceleration vector in tangential and normal components, define the motion of a particle as shown in the figure.

3 Kinematics of particles II: Curvilinear Motion

Tangential and Normal Components



- $\hat{e}_t$ and $\hat{e}'_t$ are tangential unit vectors for the particle path at P and P'.
When drawn with respect to the same origin, $\Delta \hat{e}_t = \hat{e}'_t - \hat{e}_t$ and $\Delta \theta$ is the angle between them.



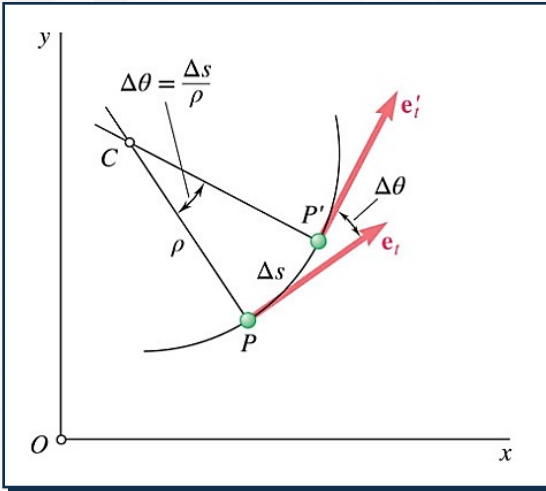
$$\Delta e_t = 2 \sin(\Delta \theta / 2)$$

$$\lim_{\Delta \theta \rightarrow 0} \frac{\Delta \hat{e}_t}{\Delta \theta} = \lim_{\Delta \theta \rightarrow 0} \frac{\sin(\Delta \theta / 2)}{\Delta \theta / 2} \hat{e}_n = \hat{e}_n$$

$$\hat{e}_n = \frac{d\hat{e}_t}{d\theta}$$

3 Kinematics of particles II: Curvilinear Motion

Tangential and Normal Components



- With the velocity vector expressed as $\vec{v} = v\vec{e}_t$ the particle acceleration may be written as

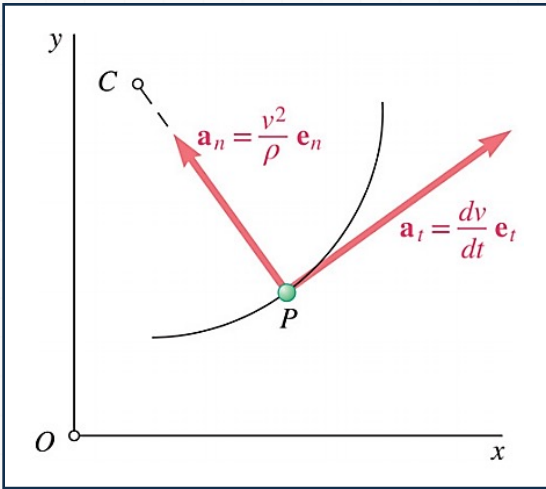
$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv}{dt}\hat{e}_t + v\frac{d\hat{e}_t}{dt} = \frac{dv}{dt}\hat{e}_t + v\frac{d\hat{e}_t}{d\theta}\frac{d\theta}{ds}\frac{ds}{dt}$$

but

$$\frac{d\hat{e}_t}{d\theta} = \hat{e}_n \quad \rho d\theta = ds \quad \frac{ds}{dt} = v$$

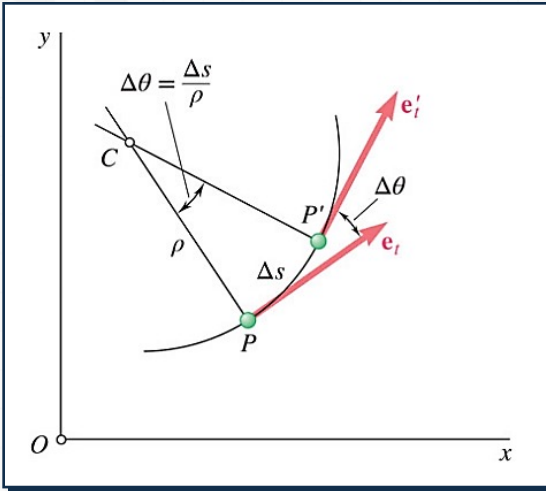
After substituting,

$$\vec{a} = \frac{dv}{dt}\hat{e}_t + \frac{v^2}{\rho}\hat{e}_n \quad a_t = \frac{dv}{dt} \quad a_n = \frac{v^2}{\rho}$$

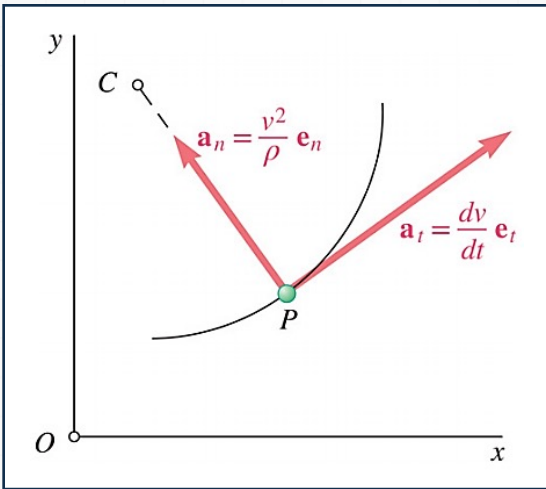


3 Kinematics of particles II: Curvilinear Motion

Tangential and Normal Components



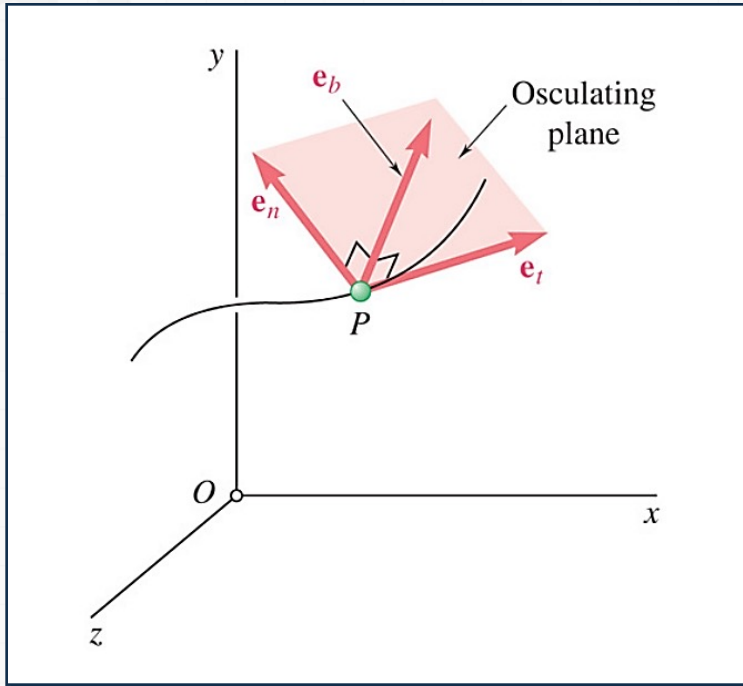
- The tangential component of acceleration reflects change of speed and the normal component reflects change of direction.



- The tangential component may be positive or negative.
Normal component always points toward center of path curvature.

3 Kinematics of particles II: Curvilinear Motion

Tangential and Normal Components



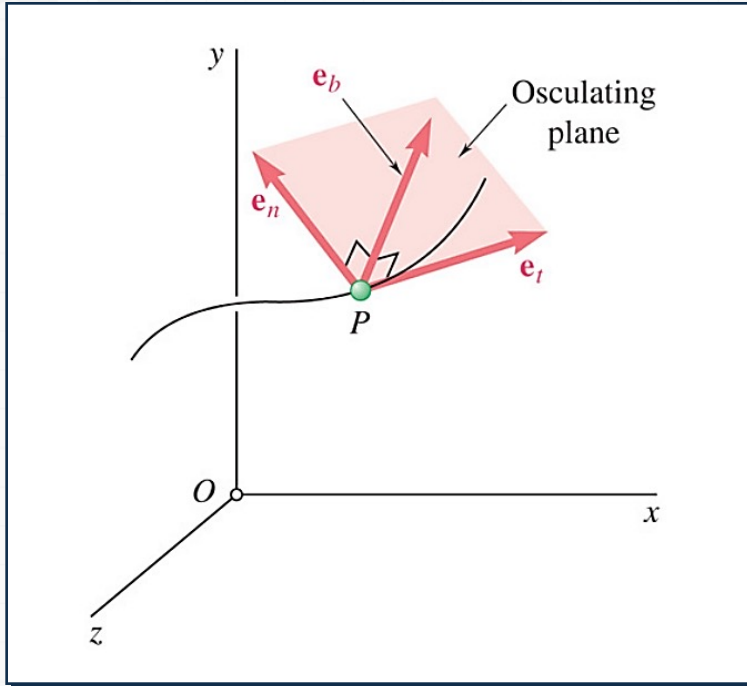
- Relations for tangential and normal acceleration also apply for particle moving along a space curve.

$$\vec{a} = \frac{dv}{dt} \hat{e}_t + \frac{v^2}{\rho} \hat{e}_n$$

$$a_t = \frac{dv}{dt} \quad a_n = \frac{v^2}{\rho}$$

3 Kinematics of particles II: Curvilinear Motion

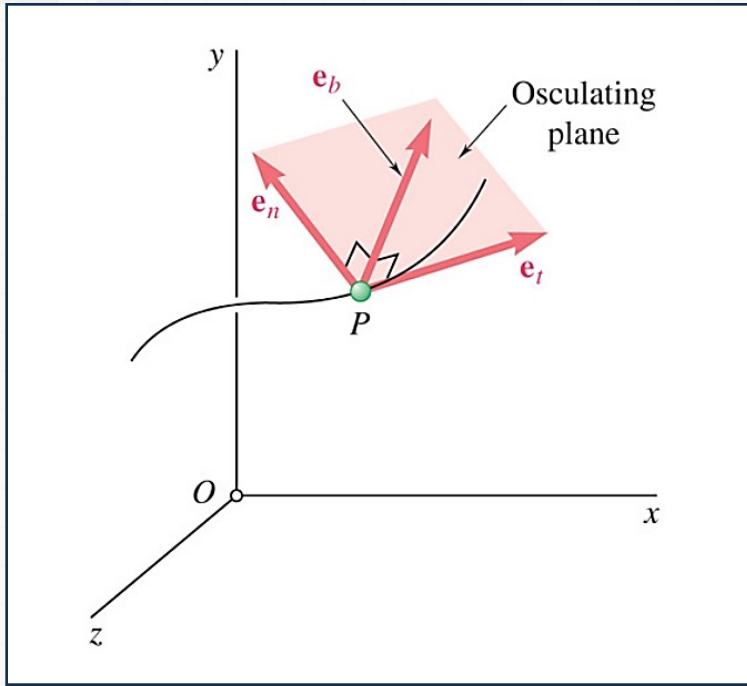
Tangential and Normal Components



- The plane containing tangential and normal unit vectors is called the *osculating plane*.

3 Kinematics of particles II: Curvilinear Motion

Tangential and Normal Components



- The normal to the osculating plane is found from

$$\hat{e}_b = \hat{e}_t \times \hat{e}_n$$

$$\hat{e}_n = \text{principal normal}$$

$$\hat{e}_b = \text{binormal}$$

- Acceleration has no component along the binormal.

3 Kinematics of particles II: Curvilinear Motion

Radial and Transverse Components

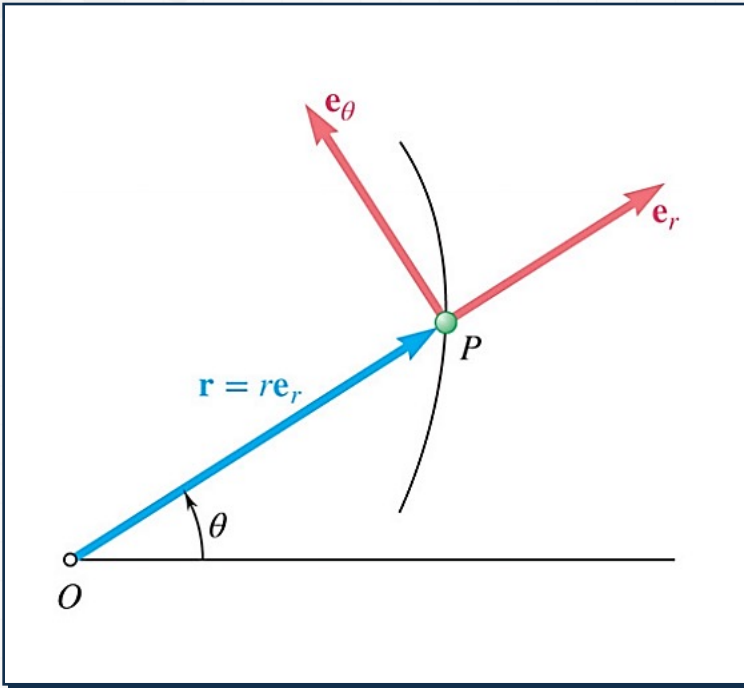
The foot pedal on an elliptical machine rotates about and extends from a central pivot point. This motion can be analyzed using radial and transverse components



Fire truck ladders can rotate as well as extend; the motion of the end of the ladder can be analyzed using radial and transverse components.

3 Kinematics of particles II: Curvilinear Motion

Radial and Transverse Components



- The position of a particle P is expressed as a distance r from the origin O to P – this defines the radial direction $\hat{e}_r$.

The transverse direction $\hat{e}_\theta$ is perpendicular to $\hat{e}_r$,

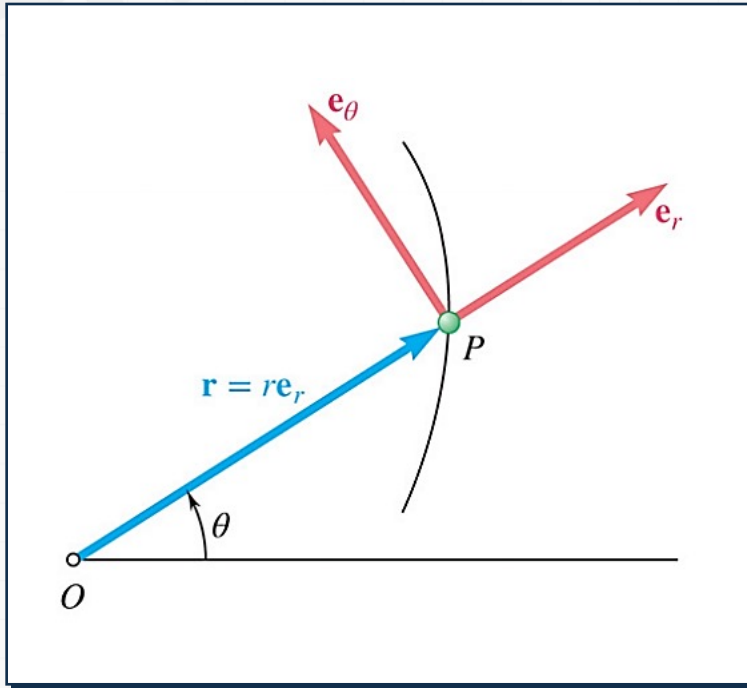
$$\vec{r} = r\hat{e}_r$$

- The particle velocity vector is:

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

3 Kinematics of particles II: Curvilinear Motion

Radial and Transverse Components

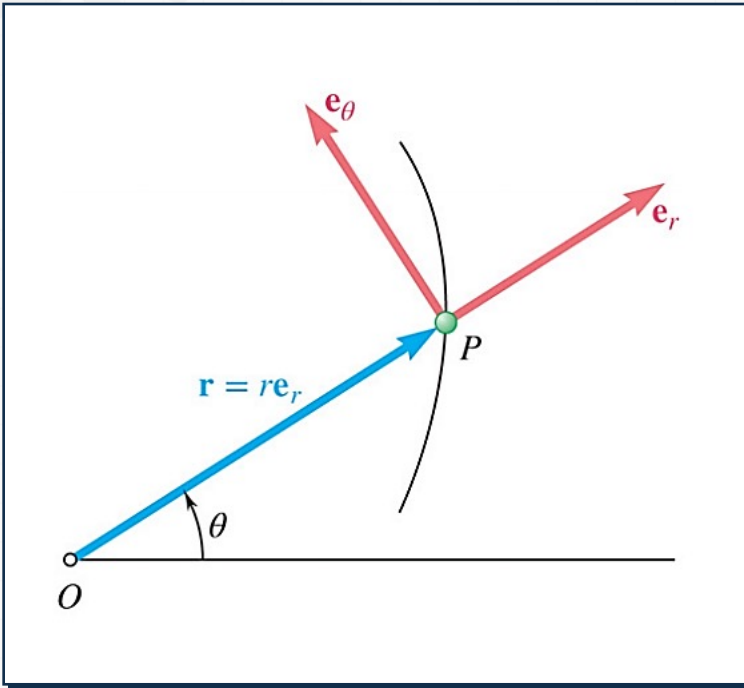


- The particle acceleration vector is:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

3 Kinematics of particles II: Curvilinear Motion

Radial and Transverse Components

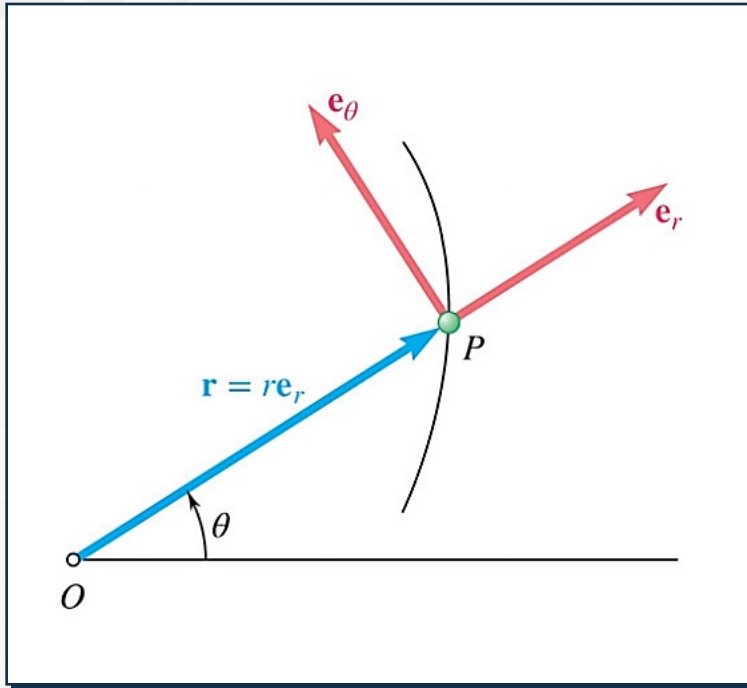


- We can derive the velocity and acceleration relationships by recognizing that the unit vectors change direction.
- The particle velocity vector is:

$$\begin{aligned}\vec{v} &= \frac{d}{dt}(r\hat{e}_r) = \frac{dr}{dt}\hat{e}_r + r\frac{d\hat{e}_r}{dt} = \dot{r}\hat{e}_r + r\frac{d\theta}{dt}\hat{e}_\theta \\ &= \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta\end{aligned}$$

3 Kinematics of particles II: Curvilinear Motion

Radial and Transverse Components



$$\vec{r} = r\hat{e}_r$$

$$\frac{d\hat{e}_r}{d\theta} = \hat{e}_\theta$$

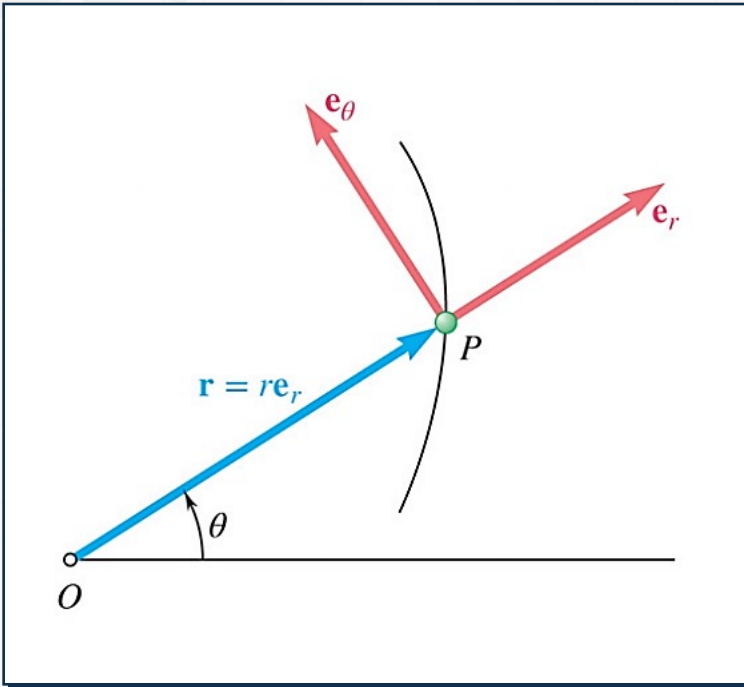
$$\frac{d\hat{e}_\theta}{d\theta} = -\hat{e}_r$$

$$\frac{d\hat{e}_r}{dt} = \frac{d\hat{e}_r}{d\theta} \frac{d\theta}{dt} = \hat{e}_\theta \frac{d\theta}{dt}$$

$$\frac{d\hat{e}_\theta}{dt} = \frac{d\hat{e}_\theta}{d\theta} \frac{d\theta}{dt} = -\hat{e}_r \frac{d\theta}{dt}$$

3 Kinematics of particles II: Curvilinear Motion

Radial and Transverse Components



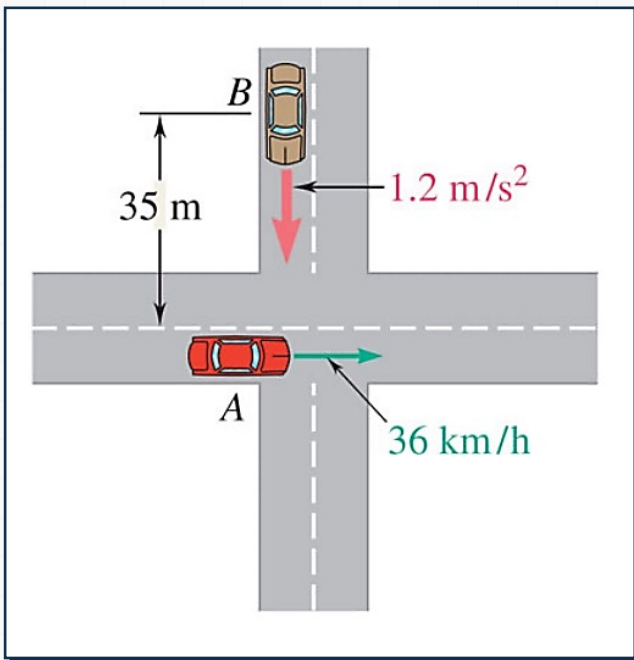
- Similarly, the particle acceleration vector is:

$$\begin{aligned}\vec{a} &= \frac{d}{dt} \left(\frac{dr}{dt} \hat{e}_r + r \frac{d\theta}{dt} \hat{e}_\theta \right) \\ &= \frac{d^2r}{dt^2} \hat{e}_r + \frac{dr}{dt} \frac{d\hat{e}_r}{dt} + \frac{dr}{dt} \frac{d\theta}{dt} \hat{e}_\theta + r \frac{d^2\theta}{dt^2} \hat{e}_\theta + r \frac{d\theta}{dt} \frac{d\hat{e}_\theta}{dt} \\ &= (\ddot{r} - r\dot{\theta}^2) \hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{e}_\theta\end{aligned}$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ Determine the position, velocity, and acceleration



- Automobile *A* is traveling east at the constant speed of 36 km/h .

As automobile *A* crosses the intersection shown, automobile *B* starts from rest 35 m north of the intersection and moves south with a constant acceleration of 1.2 m/s^2 .

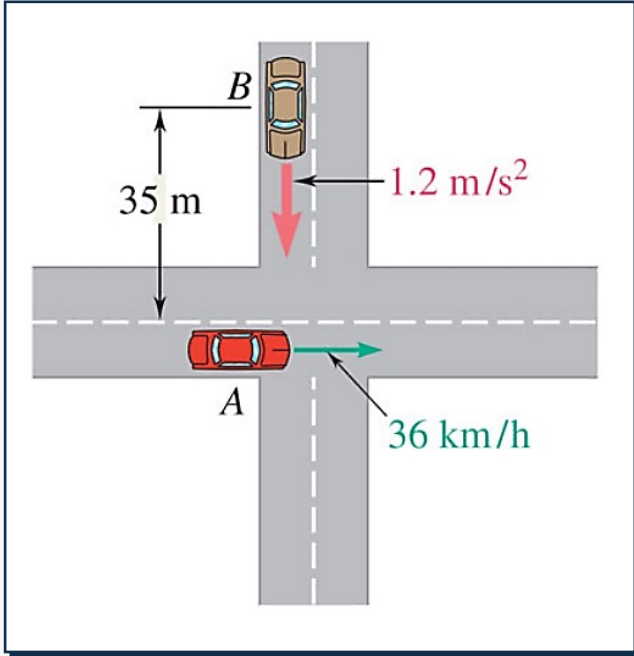
Determine the position, velocity, and acceleration of *B* relative to *A* 5 s after *A* crosses the intersection.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• **Strategy**



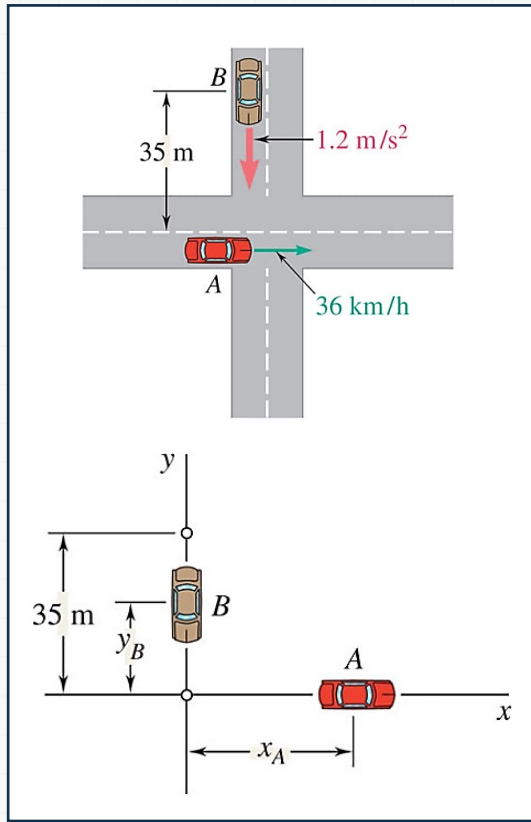
- Define inertial axes for the system.
- Determine the position, speed, and acceleration of car A at $t = 5$ s.
- Determine the position, speed, and acceleration of car B at $t = 5$ s.
- Using vectors (Equation 11.30, 11.32, and 11.33) or a graphical approach, determine the relative position, velocity, and acceleration.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• Modeling and Analysis



- Define axes along the road,

$$\text{Given: } v_A = 36 \text{ km/h}, \quad a_A = 0, \quad (x_A)_0 = 0$$

$$(v_B)_0 = 0, \quad a_B = -1.2 \text{ m/s}^2, \quad (y_B)_0 = 35 \text{ m}$$

- Determine motion of Automobile A:

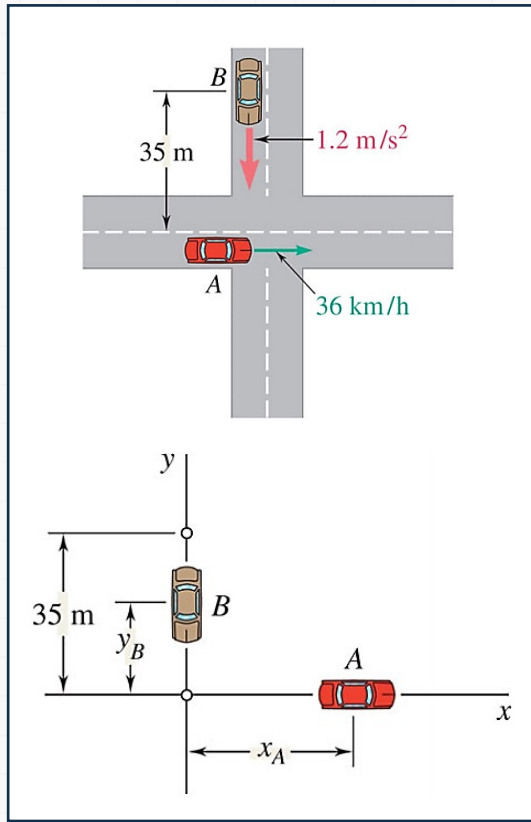
$$v_A = \left(36 \frac{\text{km}}{\text{h}} \right) \left(\frac{1000 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) = 10 \text{ m/s}$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• Modeling and Analysis



- We have uniform motion for A so:

$$a_A = 0$$

$$v_A = +10\text{m/s}$$

$$x_A = (x_A)_0 + v_A t = 0 + 10t$$

- At $t = 5\text{ s}$

$$a_A = 0$$

$$v_A = +10\text{m/s}$$

$$x_A = +(10\text{m/s})(5\text{s}) = +50\text{m}$$

$$\hat{a}_A = 0$$

$$\hat{v}_A = 10\text{m/s} \rightarrow$$

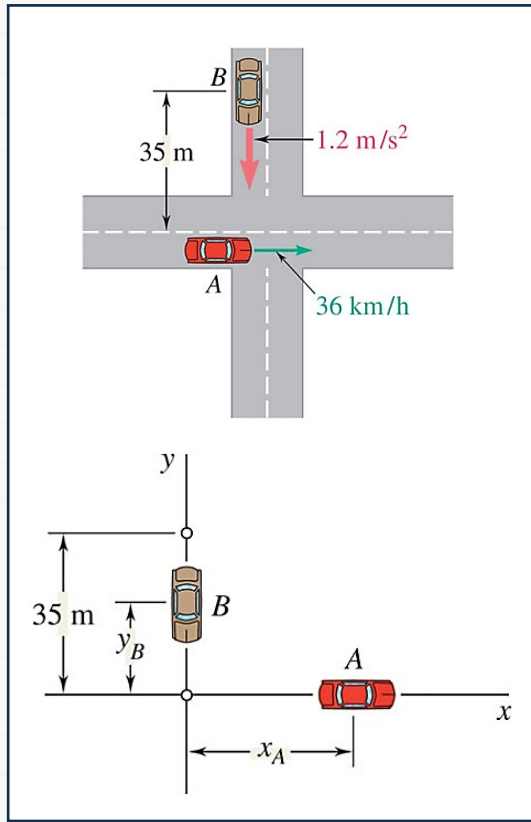
$$\hat{r}_A = 50\text{m} \rightarrow$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• Modeling and Analysis



■ Determine motion of Automobile B:

We have uniform acceleration for B so:

$$a_B = -1.2 \text{ m/s}^2$$

$$v_B = (v_B)_0 + at = 0 - 1.2t$$

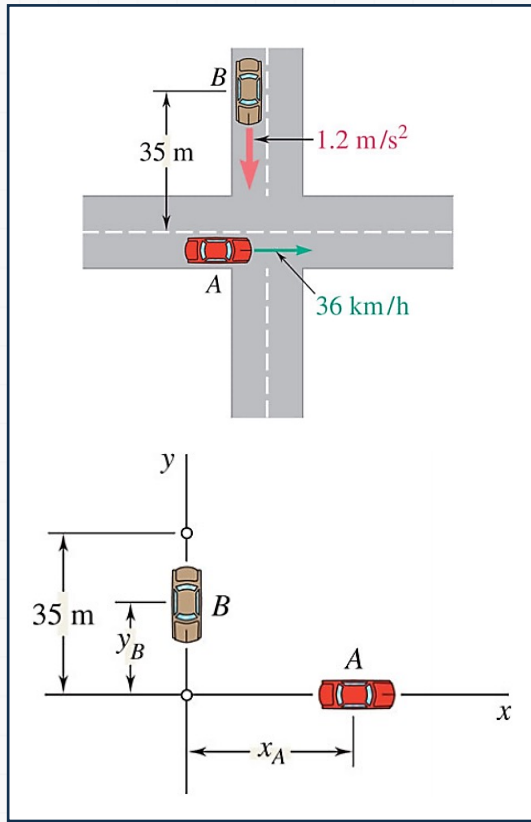
$$y_B - (y_B)_0 + (v_B)_0 t + \frac{1}{2} a_B t^2 = 35 + 0 - \frac{1}{2} (1.2) t^2$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ Determine the position, velocity, and acceleration

• Modeling and Analysis



■ At $t = 5 \text{ s}$

$$a_B = -1.2 \text{ m/s}^2$$

$$v_B = -(1.2 \text{ m/s}^2)(5 \text{ s}) = -6 \text{ m/s}$$

$$y_B = 35 - \frac{1}{2}(1.2 \text{ m/s}^2)(5 \text{ s})^2 = +20 \text{ m}$$

$$\hat{a}_B = 1.2 \text{ m/s}^2 \downarrow$$

$$\hat{v}_B = 6 \text{ m/s} \downarrow$$

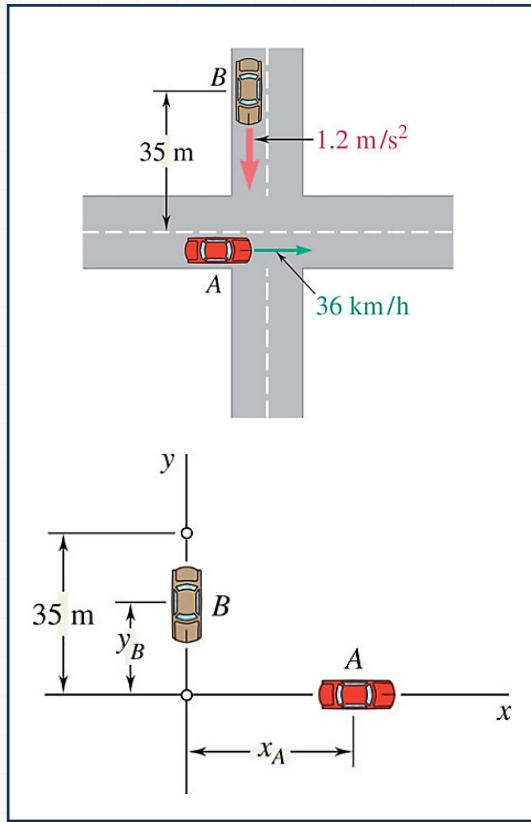
$$\hat{r}_B = 20 \text{ m} \downarrow$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• Modeling and Analysis



■ At $t = 5$ s

$$\hat{a}_A = 0$$

$$\hat{v}_A = 10 \text{ m/s} \rightarrow$$

$$\hat{r}_A = 50 \text{ m} \rightarrow$$

$$\hat{a}_B = 1.2 \text{ m/s}^2 \downarrow$$

$$\hat{v}_B = 6 \text{ m/s} \downarrow$$

$$\hat{r}_B = 20 \text{ m} \downarrow$$

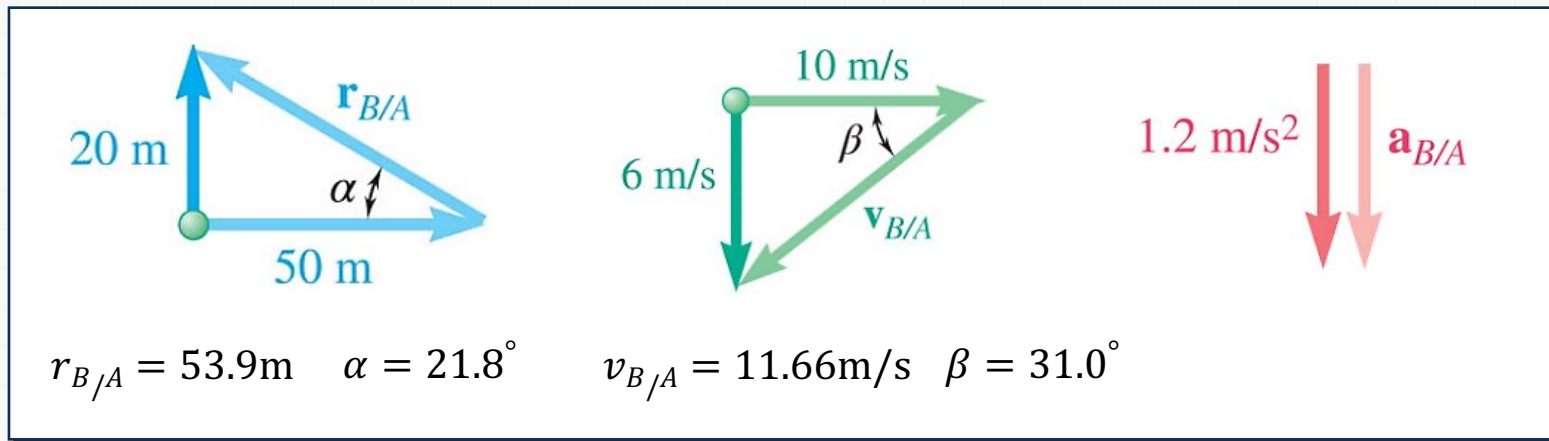
3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• **Modeling and Analysis**

- We can solve the problems geometrically, and apply the arctangent relationship:



$$\hat{\mathbf{r}}_{B/A} = 53.9\text{m} \nearrow 21.8^\circ \quad \hat{\mathbf{v}}_{B/A} = 11.66\text{m/s} \nearrow 31.0^\circ \quad \hat{\mathbf{a}}_{B/A} = 1.2\text{m/s}^2 \downarrow$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• **Modeling and Analysis**

- Or we can solve the problems using vectors to obtain equivalent results:

$$\begin{array}{lll} \hat{r}_B = \hat{r}_A + \hat{r}_{B/A} & \hat{v}_B = \hat{v}_A + \hat{v}_{B/A} & \hat{a}_B = \hat{a}_A + \hat{a}_{B/A} \\ 20\hat{j} = 50\hat{i} + \hat{r}_{B/A} & -6\hat{j} = 10\hat{i} + \hat{v}_{B/A} & -1.2\hat{j} = 0\hat{i} + \hat{a}_{B/A} \\ \hat{r}_{B/A} = 20\hat{j} - 50\hat{i}(\text{m}) & \hat{v}_{B/A} = -6\hat{j} - 10\hat{i}(\text{m/s}) & \hat{a}_{B/A} = -1.2\hat{j}(\text{m/s}^2) \end{array}$$

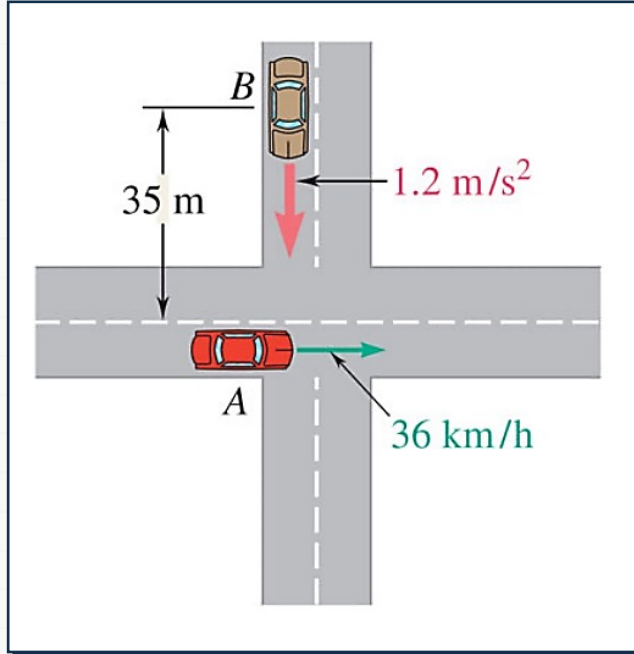
- Physically, a rider in car A would “see” car B travelling south and west.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 1

⚙️ **Determine the position, velocity, and acceleration**

• Reflect and Think

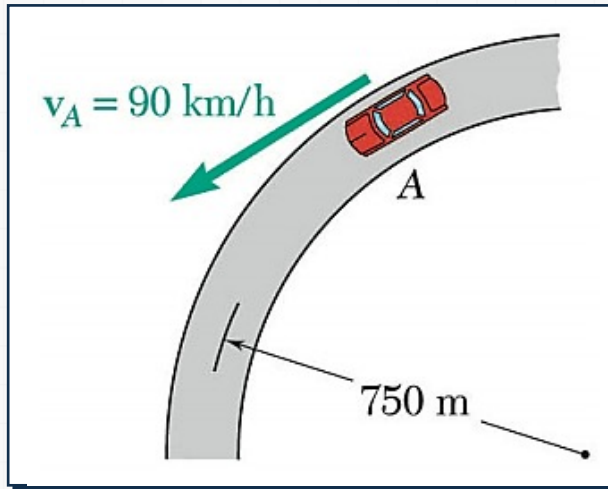


- Note that the relative position and velocity of B relative to A change with time; the values given here are only for the moment $t = 5$ s.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 2

⚙ Determine the acceleration of the automobile



- A motorist is traveling on a curved section of highway of radius 750 m at the speed of 90 km/h .

The motorist suddenly applies the brakes, causing the automobile to slow down at a constant rate.

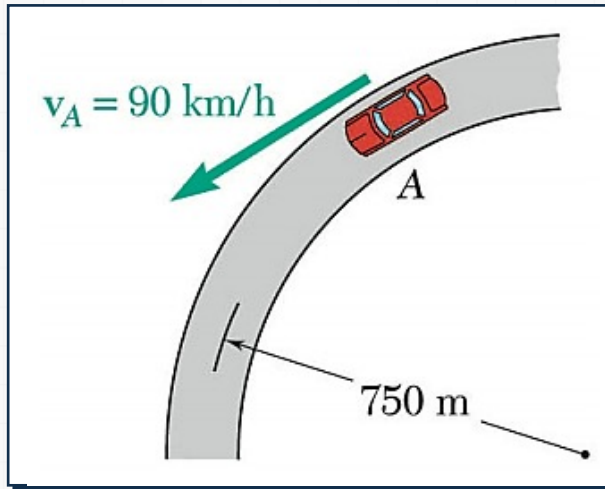
Knowing that after 8 s the speed has been reduced to 72 km/h , determine the acceleration of the automobile immediately after the brakes have been applied.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 2

⚙️ **Determine the acceleration of the automobile**

• **Strategy**



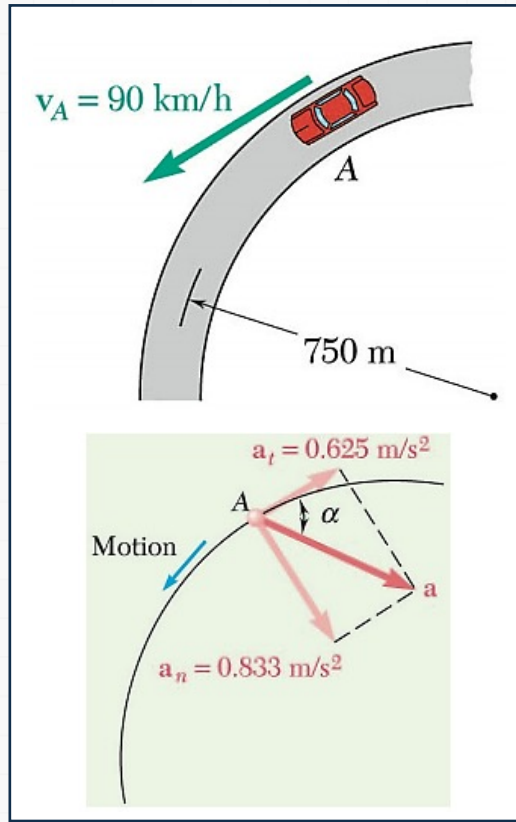
- Define your coordinate system
- Calculate the tangential velocity and tangential acceleration
- Calculate the normal acceleration
- Determine overall acceleration magnitude after the brakes have been applied

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 2

⚙ Determine the acceleration of the automobile

• Modeling and Analysis



- Define your coordinate system
- Determine velocity and acceleration in the tangential direction

$$90 \text{ km/h} = \left(90 \frac{\text{km}}{\text{h}}\right) \left(\frac{1000 \text{ m}}{1 \text{ km}}\right) \left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = 25 \text{ m/s}$$

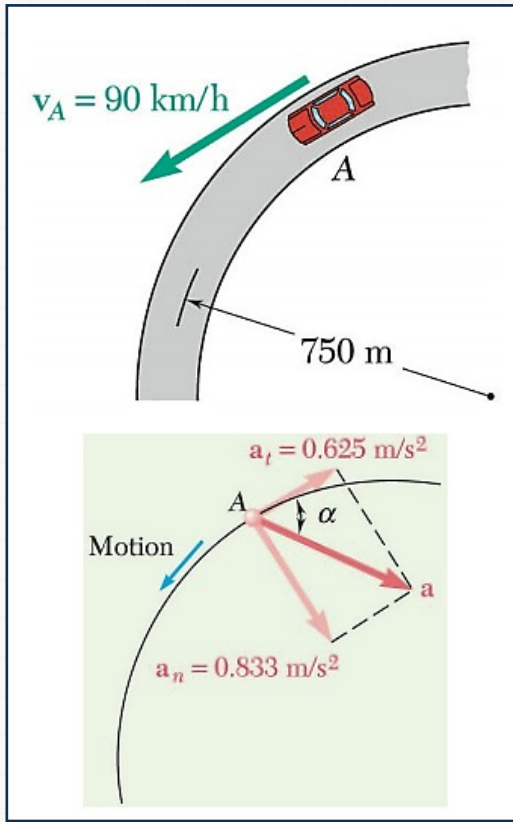
$$72 \text{ km/h} = 20 \text{ m/s}$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 2

⚙️ **Determine the acceleration of the automobile**

• Modeling and Analysis



■ The deceleration constant, therefore

$a_t = \text{average}$

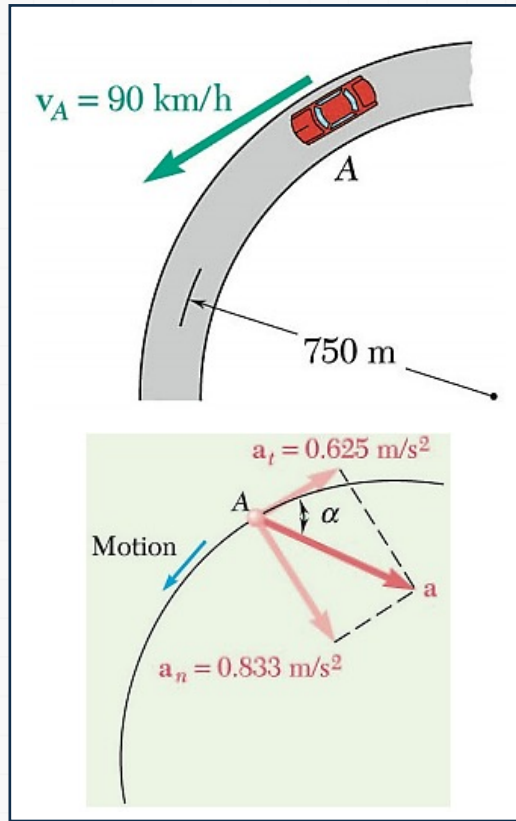
$$a_t = \frac{\Delta v}{\Delta t} = \frac{20 \text{ m/s} - 25 \text{ m/s}}{8 \text{ s}} = -0.625 \text{ m/s}^2$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 2

⚙️ **Determine the acceleration of the automobile**

• Modeling and Analysis



- Immediately after the brakes are applied, the speed is still 25 m/s

$$a_n = \frac{v^2}{\rho} = \frac{(25 \text{ m/s})^2}{750 \text{ m}} = 0.833 \text{ m/s}^2$$

$$a = \sqrt{a_n^2 + a_t^2} = \sqrt{0.625^2 + 0.833^2}$$

$$\tan \alpha = \frac{a_n}{a_t} = \frac{0.833 \text{ m/s}^2}{0.625 \text{ m/s}^2}$$

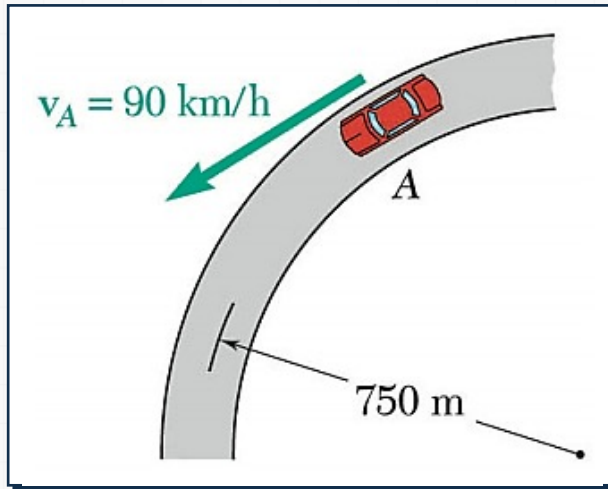
$$a = 1.041 \text{ m/s}^2 \quad \alpha = 53.1^\circ$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 2

⚙ Determine the acceleration of the automobile

• Reflect and Think



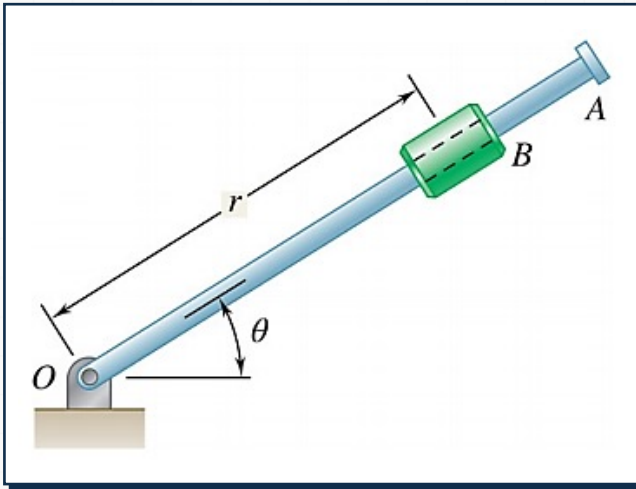
- The tangential component of acceleration is opposite the direction of motion, and the normal component of acceleration points to the center of curvature, which is what you would expect for slowing down on a curved path.

Attempting to do the problem in Cartesian coordinates is quite difficult.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

⚙️ Determine the total velocity, acceleration, and the relative acceleration



- Rotation of the arm about O is defined by $\theta = 0.15t^2$ where θ is in radians and t in seconds. Collar B slides along the arm such that $r = 0.9 - 0.12t^2$ where r is in meters.

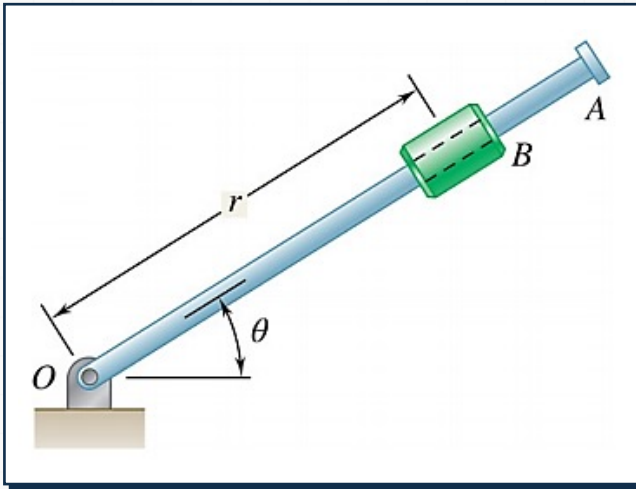
After the arm has through 30° determine
(a) the total velocity of the collar,
(b) the total acceleration of the collar, and
(c) the relative acceleration of the collar with respect to the arm.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

⚙️ **Determine the total velocity, acceleration, and the relative acceleration**

• Strategy



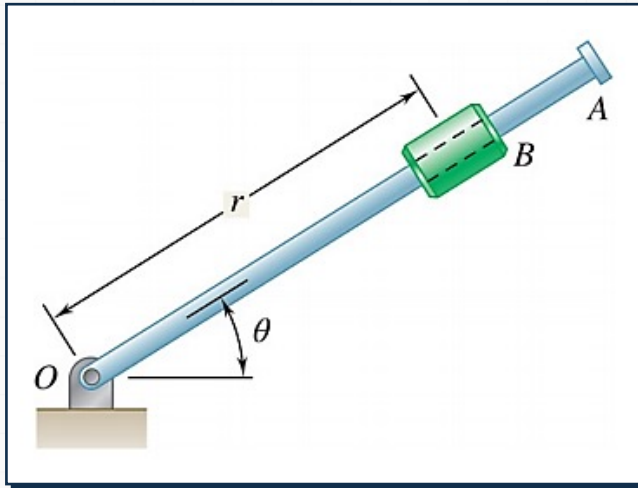
- Evaluate time t for $\theta = 30^\circ$.
- Evaluate radial and angular positions, and first and second derivatives at time t .
- Calculate velocity and acceleration in cylindrical coordinates.
- Evaluate acceleration with respect to arm.

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

⚙️ **Determine the total velocity, acceleration, and the relative acceleration**

• Modeling and Analysis



- Evaluate time t for $\theta = 30^\circ$

$$\begin{aligned}\theta &= 0.15t^2 \\ &= 30^\circ = 0.524\text{rad} \qquad t = 1.869\text{ s}\end{aligned}$$

- Evaluate radial and angular positions, and first and second derivatives at time t .

$$r = 0.9 - 0.12t^2 = 0.481\text{m}$$

$$\dot{r} = -0.24t = -0.449\text{m/s}$$

$$\ddot{r} = -0.24\text{m/s}^2$$

$$\theta = 0.15t^2 = 0.524\text{rad}$$

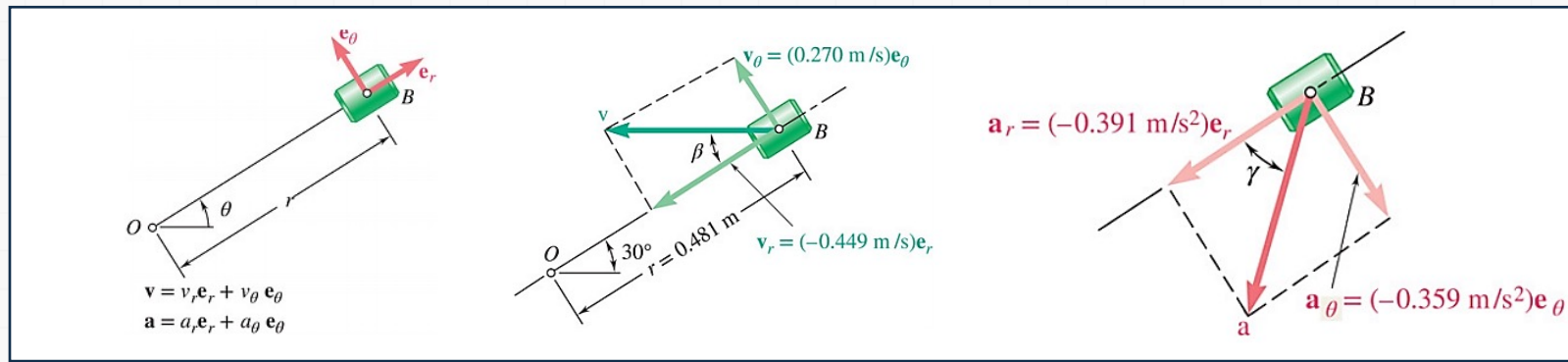
$$\dot{\theta} = 0.30t = 0.561\text{rad/s}$$

$$\ddot{\theta} = 0.30\text{rad/s}^2$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

- ⚙ Determine the total velocity, acceleration, and the relative acceleration
- Modeling and Analysis



- Calculate velocity and acceleration.

$$v_r = \dot{r} = -0.449 \text{ m/s}$$

$$v_\theta = r\dot{\theta} = (0.481 \text{ m})(0.561 \text{ rad/s}) = 0.270 \text{ m/s}$$

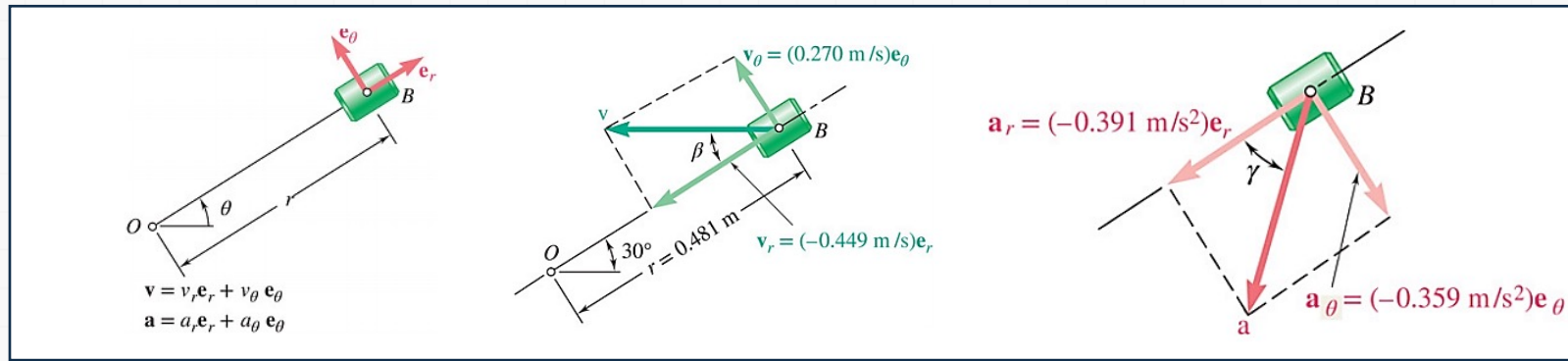
$$v = \sqrt{v_r^2 + v_\theta^2} \quad \beta = \tan^{-1} \frac{v_\theta}{v_r}$$

$$v = 0.524 \text{ m/s} \quad \beta = 31.0^\circ$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

- ⚙️ Determine the total velocity, acceleration, and the relative acceleration
- Modeling and Analysis



$$\begin{aligned}
 a_r &= \ddot{r} - r\dot{\theta}^2 \\
 &= 0.240 \text{ m/s}^2 - (0.481 \text{ m})(0.561 \text{ rad/s})^2 \\
 &= -0.391 \text{ m/s}^2
 \end{aligned}$$

$$a = \sqrt{a_r^2 + a_\theta^2} \quad \gamma = \tan^{-1} \frac{a_\theta}{a_r}$$

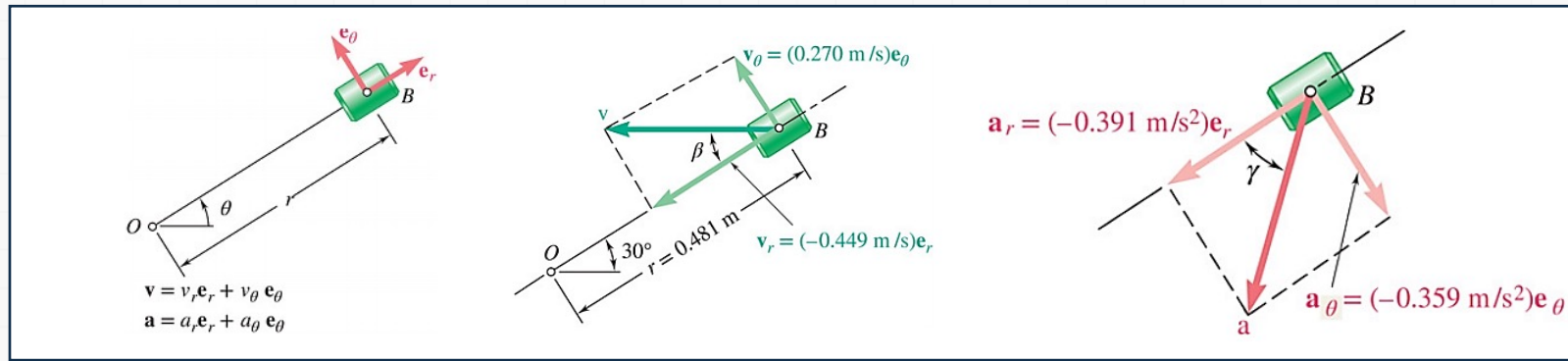
$$\begin{aligned}
 a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} \\
 &= (0.481 \text{ m})(0.3 \text{ rad/s}^2) \\
 &\quad + 2(-0.449 \text{ m/s})(0.561 \text{ rad/s}) \\
 &= -0.359 \text{ m/s}^2
 \end{aligned}$$

$$\alpha = 0.531 \text{ m/s} \quad \gamma = 42.6^\circ$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

- ⚙ Determine the total velocity, acceleration, and the relative acceleration
- Modeling and Analysis



- Evaluate acceleration with respect to arm.
Motion of collar with respect to arm is rectilinear and defined by coordinate r .

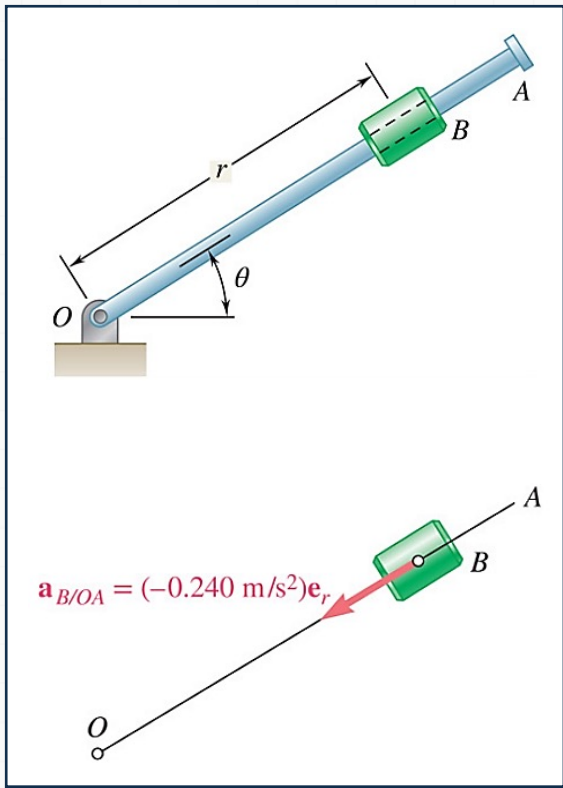
$$a_{B/OA} = \ddot{r} = -0.240 \text{ m/s}^2$$

3 Kinematics of particles II: Curvilinear Motion

Sample Problem 3

⚙️ **Determine the total velocity, acceleration, and the relative acceleration**

🔵 Reflect and Think



- You should consider polar coordinates for any kind of rotational motion. They turn this problem into a straightforward solution, whereas any other coordinate system would make this problem much more difficult.

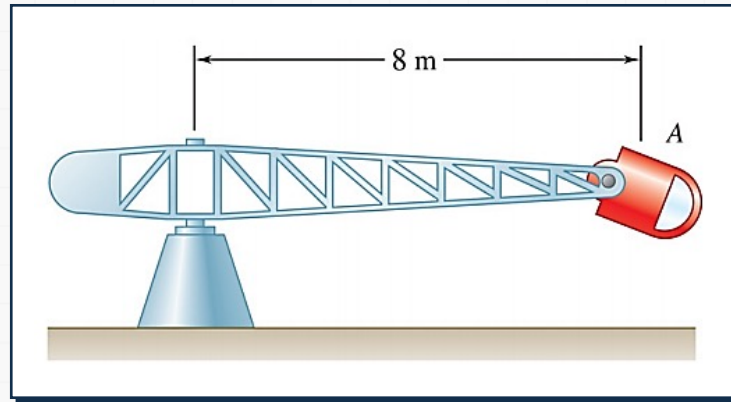
One way to make this problem harder would be to ask you to find the radius of curvature in addition to the velocity and acceleration.

To do this, you would have to find the normal component of the acceleration; that is, the component of acceleration that is perpendicular to the tangential direction defined by the velocity vector.

3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙ Determine the acceleration magnitude



- The angular acceleration of the centrifuge arm varies according to

$$\ddot{\theta} = 0.05\theta \text{ (rad/s}^2\text{)}$$

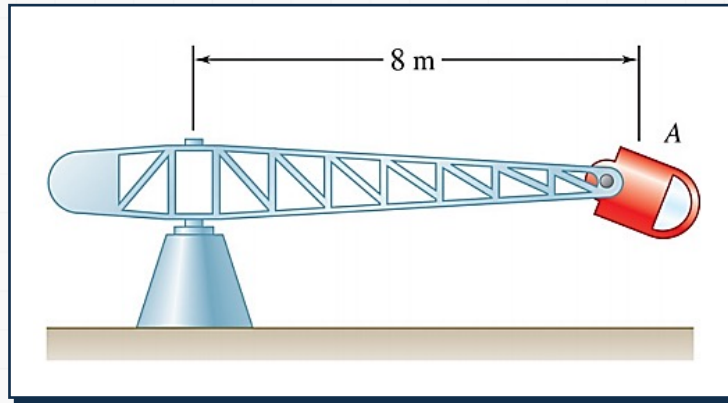
Where θ is measured in radians. If the centrifuge starts from rest, determine the acceleration magnitude after the gondola has travelled two full rotations.

3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙ Determine the acceleration magnitude

• Strategy



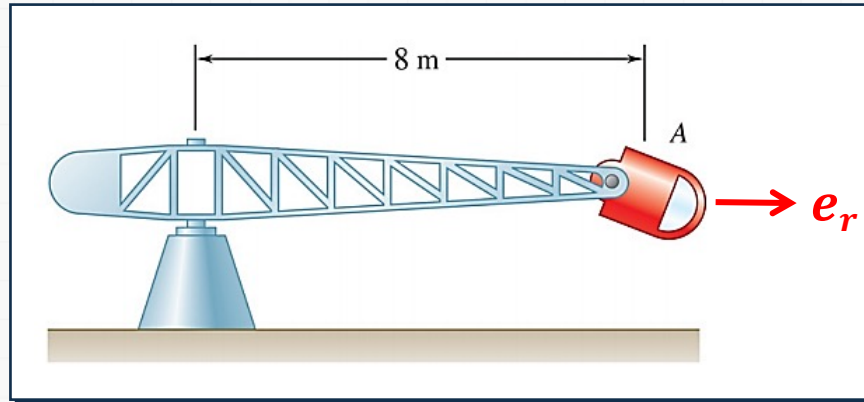
- Define your coordinate system.
- Calculate the angular velocity after three revolutions.
- Calculate the radial and transverse accelerations.
- Determine overall acceleration magnitude.

3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙ Modeling and Analysis

- Define your coordinate system
 - In the side view, the transverse direction points into the “page”



3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

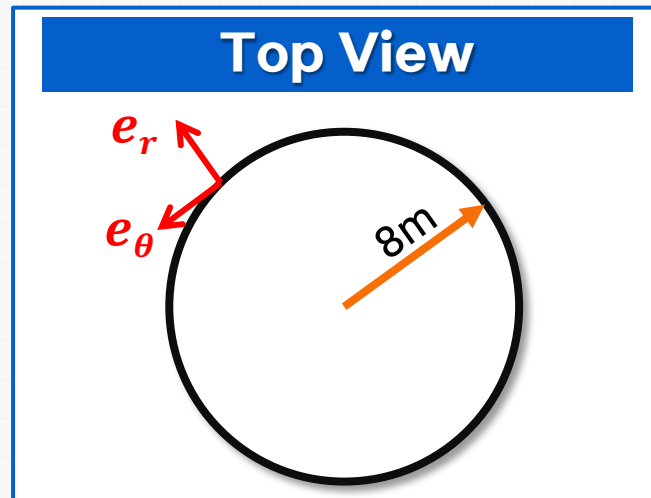
⚙️ Modeling and Analysis

- Define the angular velocity

$$\ddot{\theta} = 0.05\theta \text{ (rad/s}^2\text{)}$$

- Acceleration is a function of position, so use:

$$\ddot{\theta}d\theta = \dot{\theta}d\dot{\theta}$$



3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙️ Modeling and Analysis

- Evaluate the integral

$$\int_0^{(2)(2\pi)} 0.05 \theta d\theta = \int_0^{\dot{\theta}} \dot{\theta} d\dot{\theta}$$

$$\frac{0.05\theta^2}{2} \Big|_0^{2(2\pi)} = \frac{\dot{\theta}^2}{2} \Big|_0^{\dot{\theta}}$$

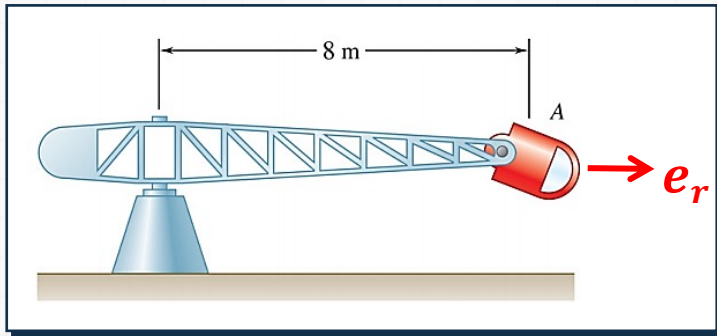
$$\dot{\theta}^2 = 0.05[2(2\pi)]^2$$

3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙ Modeling and Analysis

- Determine the angular velocity



$$\dot{\theta}^2 = 0.05[2(2\pi)]^2$$

$$\dot{\theta} = 2.8099 \text{ rad/s}$$

- Determine the angular acceleration

$$\ddot{\theta} = 0.05\dot{\theta} = 0.05(2.8099) = 0.1405 \text{ rad/s}^2$$

3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙️ Modeling and Analysis

- Find the radial and transverse accelerations

$$\begin{aligned}\vec{a} &= (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta \\ &= (0 - (8)(2.8099)^2)\hat{e}_r + ((8)(0.6283) + 0)\hat{e}_\theta \\ &= -63.166\hat{e}_r + 5.0265\hat{e}_\theta (\text{m/s}^2)\end{aligned}$$

- Magnitude

$$a_{mag} = \sqrt{a_r^2 + a_\theta^2} = \sqrt{(-63.166)^2 + [5.0265]^2}$$

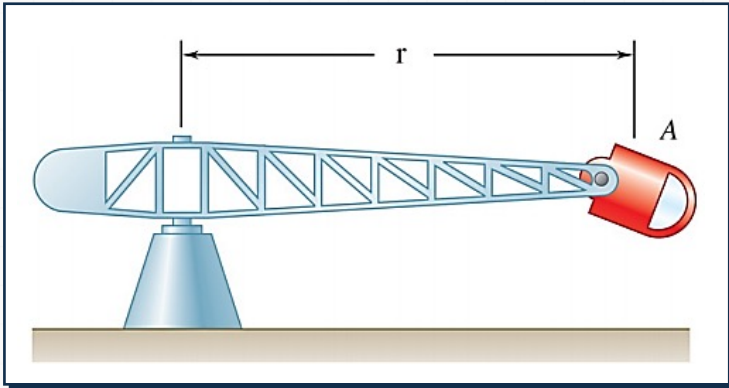
$$a_{mag} = 63.365 \text{ m/s}^2$$

3 Kinematics of particles II: Curvilinear Motion

Group Problem Solving

⚙️ Reflect and Think

- What would happen if you designed the centrifuge so that the arm could extend from 6 to 10 meters?



- You could now have additional acceleration terms. This might give you more control over how quickly the acceleration of the gondola changes (this is known as the G-onset rate).

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

3 Kinematics of particles II: Curvilinear Motion

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 11